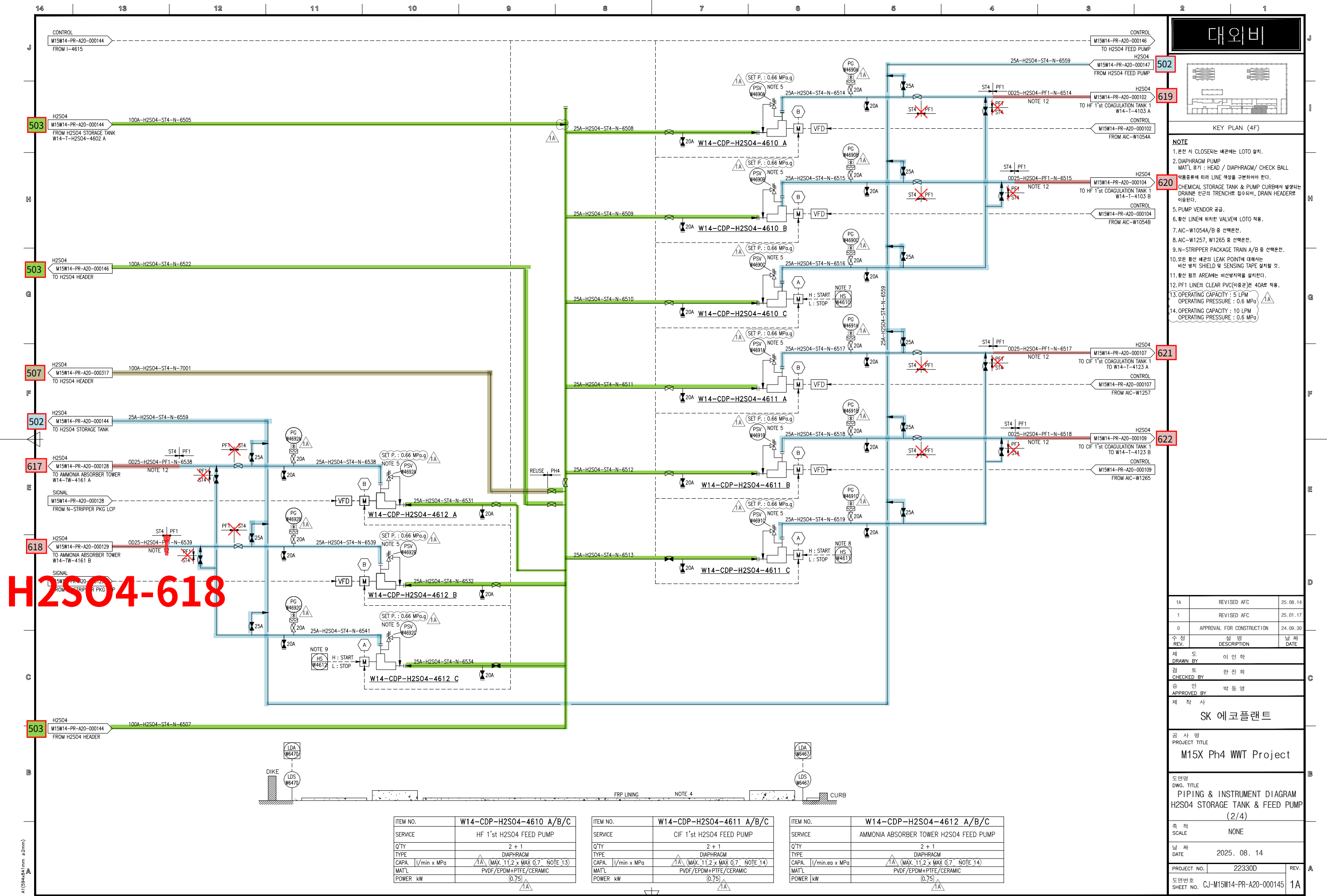


DRAWING LIST FOR TEST PACKAGE PROJECT NO.: 22328D PROJECT TITLE : M15X Ph2 Project		H2SO4-618
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PROJECT NO.: 22328D**PROJECT TITLE : M15X Ph2 Project**

H2SO4-618

[illegible]



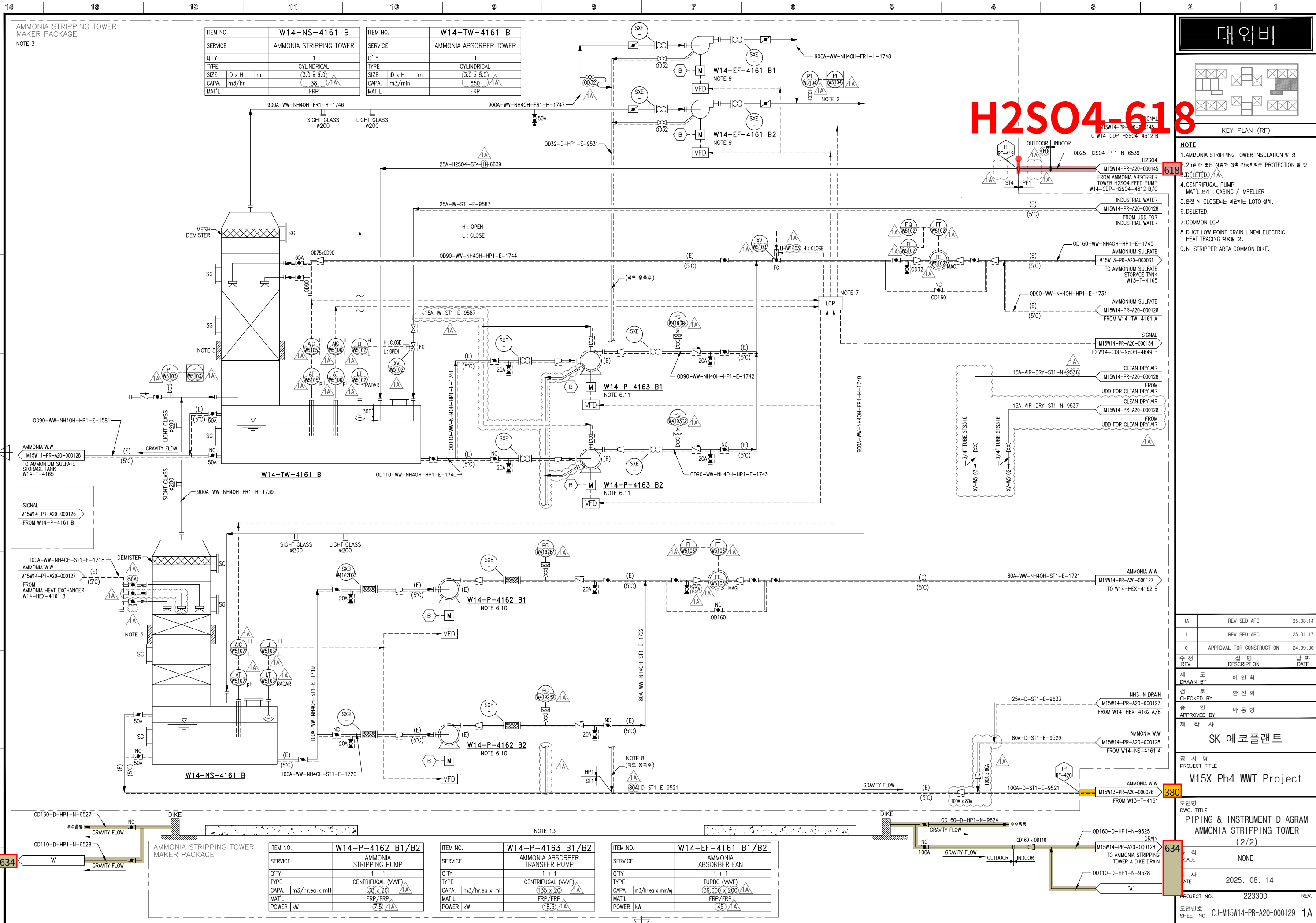
대외비

KEY PLAN (4F)

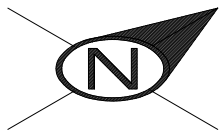
NOTE

1. 운전 시 CLOSE되는 배관에는 LOTO 설치.
2. DIAPHRAGM PUMP MAT'L 표기 : HEAD / DIAPHRAGM/ CHECK BALL.
3. 역류종류에 따라 LINE 색상을 구분하여야 한다.
4. CHEMICAL STORAGE TANK & PUMP CURB에서 발생하는 DRAIN은 인근의 TRENCH로 침수되어, DRAIN HEADER로 이송된다.
5. PUMP VENDOR 공급.
6. 황산 LINE에 위치한 VALVE에 LOTO 적용.
7. AIC-W1054A/B 중 선택운전.
8. AIC-W1257, W1265 중 선택운전.
9. N-STRIPPER PACKAGE TRAIN A/B 중 선택운전.
10. 모든 황산 배관의 LEAK POINT에 대해서는 비산 방지 SHIELD 및 SENSING TAPE 설치할 것.
11. 황산 침트 AREA에는 비산방지막을 설치한다.
12. PF1 LINE의 CLEAR PVC(이동관)은 40A로 적용.
13. OPERATING CAPACITY : 5 LPM
OPERATING PRESSURE : 0.6 MPa
14. OPERATING CAPACITY : 10 LPM
OPERATING PRESSURE : 0.6 MPa

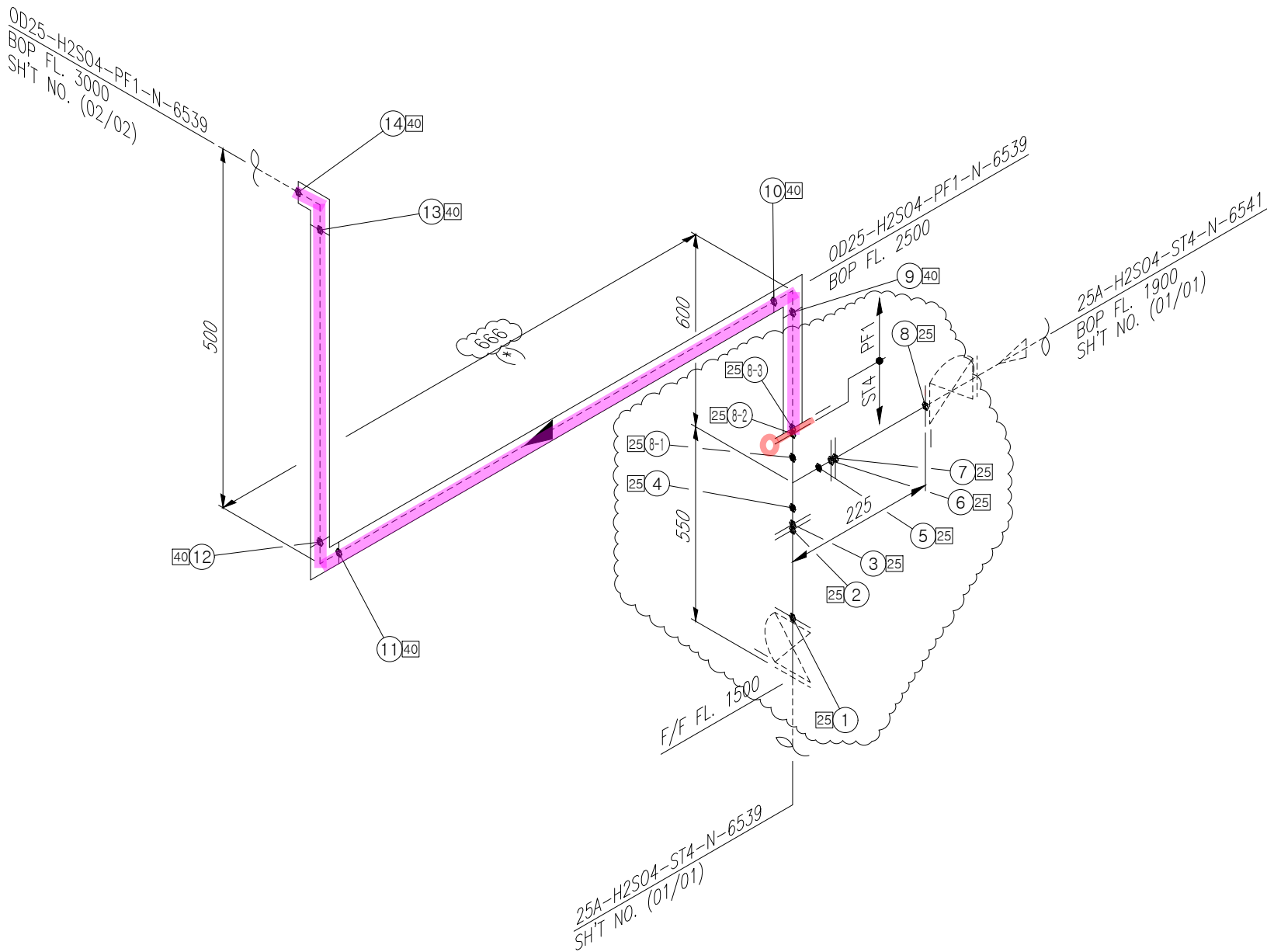
1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.09.30
수정	실명	날짜
REV.	DESCRIPTION	DATE
제	도	이 인 학
DRAWN BY		
검	토	한 진 희
CHECKED BY		
승	인	박 동 영
APPROVED BY		
제	작	사
SK 에코플랜트		
공 사 명		
PROJECT TITLE		
M15X Ph4 WWT Project		
도면명		
DWG. TITLE		
PIPING & INSTRUMENT DIAGRAM		
H2SO4 STORAGE TANK & FEED PUMP		
(2/4)		
속		
척		
SCALE		
NONE		
날		
짜		
DATE		
2025. 08. 14		
PROJECT NO.		22330D
도면번호		SHEET NO. CJ-M15W14-PR-A20-000145
R1		1A



대외비		
KEY PLAN (RF)		
NOTE		
1. AMMONIA STRIPPING TOWER INSULATION 및 2. 2m이하 또는 사람과 접촉 가능지역은 PROTECTION 할 것		
3. DELETED		
4. CENTRIFUGAL PUMP MAT'L 표기 : CASING / IMPELLER		
5. 로트 시 CLOSER는 배관에는 LOTO 설치.		
6. DELETED.		
7. COMMON LCP.		
8. DUCT LOW POINT DRAIN LINE에 ELECTRIC HEAT TRACING 적용할 것.		
9. N-STRIPPER AREA COMMON DIKE.		
1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.09.30
수정	실명	날짜
REV.	DESCRIPTION	DATE
제 도 이 인 학		
검 토 한 진 회		
승 인 박 동 영		
제 작 사		
SK 에코플랜트		
공 사 명		
PROJECT TITLE		
M15X Ph4 WWT Project		
도면명		
DWG. TITLE		
PIPING & INSTRUMENT DIAGRAM		
AMMONIA STRIPPING TOWER		
(2/2)		
척		
SCALE		
NONE		
제		
DATE		
2025. 08. 14		
PROJECT NO.		
22330D		
REV.		
도면번호		
SHEET NO. CJ-M15W14-PR-A20-000129		
1A		



2



<NOTE>
1. TRAY & ANGLE,CHANNEL SUPPORT가 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST



ITEM	SIZE	Q'TY	DESCRIPTION
PF1			
P	OD25	1.8M	PIPE, PFA TUBE, PE, FERRULE, 1.57MM TH'K
P3	40A	1.8M	PIPE, CLEAR PVC, PE, SOCKET BONDING, 4.0MM TH'K
E90A	40A	3EA	ELBOW 90 DEG, CLEAR PVC, PE, SOCKET BONDING, 4.0MM TH'K
F1	OD25	1EA	FLG, PFA STUB END, BACK RING(PTFE), LAP JOINT(STS 304)

ST4			
P	25A	0.7M	PIPE, STS 304TP+PTFE Lined, ERW - SCH 10S
T	25A	1EA	TEE EQUAL, STSF304+PTFE Lined, ERW - SCH 10S
F	25A	4EA	FLG SO, KS 10K, RF, STS304+PTFE Lined
G	25A	4EA	GASKET, KS 10K, RF, PTFE, 3.0MM TH'K
B1	25A	4SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
SH	25A	3EA	SHIELD

CLIENT

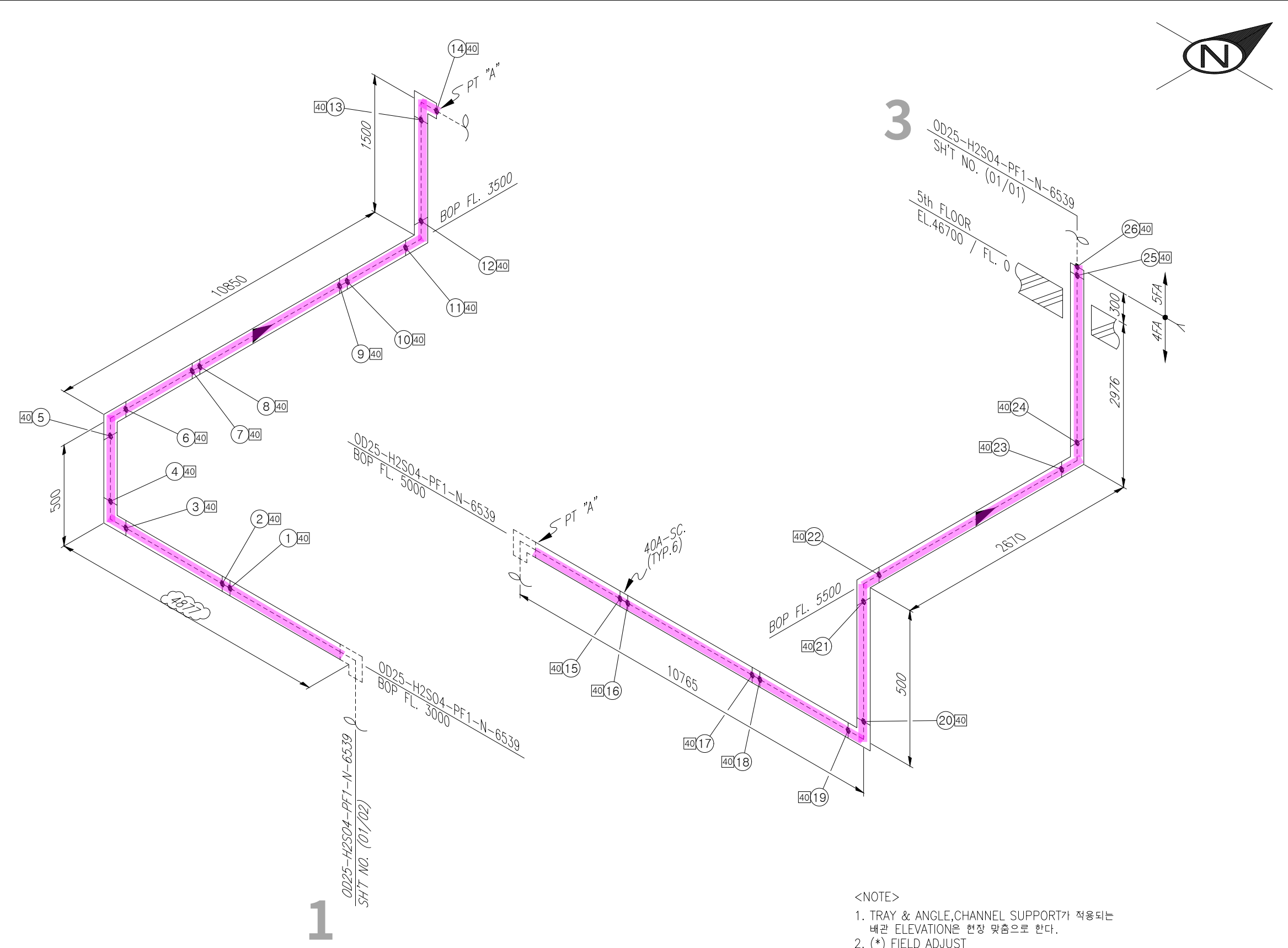


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CONTRACTOR		PROJECT NAME	
SK ecoplant		M15X Ph4 Project	
COMMODITY		JOB NO.	
H2SO4		22330D	
LINE NO.		SH'T NO.	REV.
OD25-H2SO4-PF1-N-6539		01 / 02	0

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

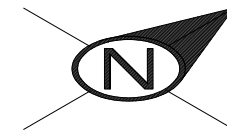
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14												
	THICK.	-	DESIGN	6.7	60	UNIT	4FA												
			OPERATING	6.1	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040013-A													
						REF. P & ID CJ-M15W14-PR-A20-000145													
								△	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영						
								REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D						



<NOTE>
1. TRAY & ANGLE,CHANNEL SUPPORT가 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PIPE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14								CONTRACTOR	PROJECT NAME	
	THICK.	-	DESIGN	6.9	60	UNIT	4FA								M15X Ph4 Project		
			OPERATING	6.2	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040008-A								COMMODITY	H2SO4	JOB NO. 22330D	
						REF. P & ID CJ-M15W14-PR-A20-000145			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO.	OD25-H2SO4-PF1-N-6539	SH'T NO. 02 / 02	REV. 0
								REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D				



1. (*) FIELD ADJUST
2. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.

CLIENT

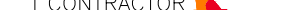


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경고 : 본 문건은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 무단으로 복제, 제3자에게 공개, 누설 또는 사용함으로써 회사에 손해가 발생할 경우 응징 조치될 수 있으며, 회사는 응징 조치를 할 권리가 있습니다.

[illegible]

CLIENT		 4	
CONTRACTOR 		PROJECT NAME M15X Ph4 Project	
COMMODITY H2S04		JOB NO. 22330D	
LINE NO. 0D25-H2S04-PF1-E-6539		SH'T NO. 01 / 01	REV. 1

INSULATION	CODE	E	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME		
	THICK.	25	DESIGN	6.9	60	UNIT	RFA								M15X Ph4 Project		
			OPERATING	6.2	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-R10005-A			25.03.31	REVISED AS MARKED	황래중	안남후	박동영	COMMODITY	H2S04	JOB NO.	
						REF. P & ID CJ-M15W14-PR-A20-000129			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영			22330D	
								REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	OD25-H2S04-PF1-E-6539	SH'T NO. 01 / 01	REV. 1